Internal Assessment Question Paper – 2

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of CSE

Programme: B.E., CSE.
CIE:2
Date: 27/02/2024

Term: Nov 2023-Feb 2024
Course Code: CS34

Sem: 3
Course: Object Oriented Programming
Max Marks: 30
Time: 03:00 pm to 04:00 pm

Instructions to Candidates: 1. **Question 1 is compulsory**
2. Answer any one full question from 2 or 3. Each Question carries 15M.
3. Programmable Calculators, Mobile, Smart watches, or any Electronic gadgets/Test Text material in any form are strictly banned.

Sl#	Question	Marks	Bloom's Level	CO Mapping
1 a	Explain try, catch, finally with syntax and examples.	05	Apply	CO4
1b	Write a Java Program to implement Packages by performing following operations: a. Create a class **TwoDim** which contains private members as **x** and **y** coordinates in package **P1**. Define the default constructor, a parameterized constructor and override **toString()** method to display the coordinates. b. Reuse the class **TwoDim** and in package **P2** create another class **ThreeDim**, adding a new dimension as **z** as its private member. Define the constructors for the subclass and override the **toString()** method in the subclass also. c. Write a driver code that imports both packages and usage of classes **TwoDim** and **ThreeDim** by creating objects.	05	Apply	CO3
1c	Explain Delegation Event Model in Java.	05	Understand	CO5
2 a	Explain Lambda Expression in Java with syntax. Write a Java program to implement "ADDITION" and "MULTIPLICATION" of two numbers using Lambda Expressions.	08	Apply	CO5
2b	Write a program to create two threads t1, t2 which should print odd numbers, and reverse of a number respectively and stop the thread after creating 3 odd numbers.	07	Apply	CO4

PTO

3a	Design an Event handling Program to implement three buttons "MSRIT" , "PESU" and "RVCE" to display a message "Welcome to MSRIT" , "Welcome to PESU" and Welcome to "RVCE" when each button is clicked. Each object should be able to register to event listeners. When an event occurs, the program should notify the registered event listeners for that event type.	08	Apply	CO5
3b	Write a Java program to find the area of a triangle with three sides a, b, c. A triangle can be formed only if a+b>c, b+c>a, c+a>b. First verify whether the above three conditions are satisfied. If any one of them is not satisfied then throw an user defined exception called ValidateTriangle Exception	07	Apply	CO3

Course Outcomes (COs):

CO3: Develop java programs using inheritance, interfaces and packages. (PO-2,3,5PSO-3)

CO4: Explore the exception handling mechanism and thread synchronization. (PO-2,3,5PSO-3)

CO5: Design the GUI application using swings and handle the interactions. (PO-2,3,5PSO3)

Faculty Coordinator: Jamuna S Murthy

Jamuna

27/2/24

HOD,CSE

Internal Assessment Question Paper – 1

Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of CSE

Programme: B.E., CSE.
CIE:1
Term: Nov 2023-Feb 2024
Course Code: CS34
Sem: 3
Course: Object Oriented Programming

Portions for Test: L1-L28
Date: 06/01/2024
Max Marks: 30
Time: 09:30 am to 10:30 am

Instructions to Candidates: 1. **Question 1 is compulsory**
2. Answer any one full question from 2 or 3. Each Question carries 15M.
3. Programmable Calculators, Mobile, Smart watches, or any Electronic gadgets/Test Text material in any form are strictly banned.

Sl#	Question	Marks	Bloom's Level	CO Mapping
1 a	Explain the below with an example each: a. this keyword b. finalize c. super d. final e. Garbage Collection in java	07	Understand	CO3
1b	Differentiate between Object-Oriented Design and Object- Oriented Analysis.	03	Understand	CO1
1c	Consider a class hierarchy consisting of three classes: Vehicle, Car, and ElectricCar. Each class has the following methods: start: Prints a message indicating that the vehicle is starting. stop: Prints a message indicating that the vehicle is stopping. The Car class inherits from Vehicle, and the ElectricCar class inherits from Car. The ElectricCar class also has an additional method: charge: Prints a message indicating that the electric car is charging. write the Java code for these classes and demonstrate their use in a program. Create objects of each class and invoke their methods to showcase the inheritance hierarchy and the unique behavior of each class.	05	Apply	CO2
2 a	Create a Java program that models a simple banking system using method overloading. Design a class named **BankAccount** with methods for performing various banking operations. Implement multiple versions of the transfer method to support transferring funds between accounts for different scenarios, such as transferring from one account to another, transferring to a specific	08	Apply	CO3

	account number, and transferring a specific amount to multiple accounts. In the main method of another class, instantiate multiple **BankAccount** objects and demonstrate the usage of each version of the transfer method. Discuss how method overloading in this banking example contributes to the flexibility and maintainability of the code.			
2b	Analyze the Object-Oriented Programming (OOP) principles - encapsulation, inheritance, and polymorphism - in the context of Java programming.	07	Analyze	CO1
3 a	Instantiate several **Employee** objects and assign them to different departments within a Company. Display the total number of employees and total salary expenses. Discuss the significance of static variables in the **Employee** class. How do they help in maintaining a count of total employees and tracking salary expenses? Implement methods in the **Employee** class to display the total number of employees and total salary expenses.	08	Apply	CO2
3c	Explain Java Buzz words in detail.	07	Understand	CO1

CIE: Internal Assessment Details

Internal Assessment Question Paper – 1
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

Programme: B.E (CSE)	**Term:** Nov 2023 – Mar 2024	**Max Marks:** 30
CIE: Test 1	**Subject:** Discrete Mathematical Structures	**Sub Code:** CS35
Credits : 3:0:0	**Sem:** III	**Sec:** A, B, C
Date: 6/1/2024	**Time:** 12.30 - 1:30 PM	**Portions for Test:** (L1-L10)

Instructions to Candidate:

1. **Question 1 is Compulsory.** Answer any one question from 2 and 3.
2. Each Question carries 15M.
3. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl. No	Question	Marks	Bloom's Level	CO Mapping
1	(a) Establish the validity of the following argument using rules of inference. If Dominic goes to the racetrack, then Helen will be mad. If Ralph plays cards all night, then Carmela will be mad. If either Carmela or Helen gets mad then their lawyer Veronica will be notified. Veronica has not heard from either of these two clients. Therefore, Dominic did not make it to the racetrack and Ralph didn't play cards all night	05	Apply	CO1
	(b) Define Partial order on a set A. Construct Hasse Diagram for the (i) Divides relation on the set {2,4,5,10,12,20,25} (ii) Poset (D_{72},|)	04	Apply	CO2
	(c) With steps and reasons establish the validity of the argument: (i) $\forall x[p(x) \vee q(x)]$ $\exists x \neg p(x)$ $\forall x[\neg q(x) \vee r(x)]$ $\forall x[s(x) \rightarrow \neg r(x)]$ $\therefore \exists x \neg s(x)$ (ii) $\forall x,[p(x) \vee q(x)]$ $\forall x,[(\neg p(x) \wedge q(x)) \rightarrow r(x)]$ $\therefore \forall x,[\neg r(x) \rightarrow p(x)]$	06	Apply	CO1
2	**(a)** Establish the validity of the following Arguments: (i) $p \rightarrow q$ $q \rightarrow (r \wedge s)$ $\neg r \vee (\neg t \vee u)$ $p \wedge t$ $\therefore u$ (ii) p $p \rightarrow r$ $p \rightarrow (q \vee \sim r)$ $\sim q \vee \sim s$ $\therefore s$	06	Apply	CO1
	(b) Let A = {1, 2, 3, 4}, B = {w, x, y, z} and C={p, q, r, s}. Consider R_1 = {(1, x), (2, w), (3, z)}, a relation from A to B, R_2= {(w, p), (z, q), (y, s), (x, p)}, a relation from B to C. i)What is the composite relation $R_1 \circ R_2$ from A to C ii) Write the relation matrices $M(R_1)$, $M(R_2)$ and $M(R_1 \circ R_2)$ iii) Verify $M(R_1) \cdot M(R_2) = M(R_1 \circ R_2)$	04	Understand	CO2

2,4

(P.T.O)

	(c) State and Verify Absorption law and Domination Law	05	Understand	CO1
3	**(a)** Define logical equivalence of two propositions. Prove the following logical equivalences using laws of logic: (i) $p \vee [p \wedge (p \vee q)] \Leftrightarrow p$ (ii) $(p \rightarrow q) \wedge (\neg q \wedge (r \vee \neg q)) \Leftrightarrow \neg[q \vee p]$	05	Apply	CO1
	(b) Determine whether each of the following relations are reflexive, symmetric and transitive: (i) Relation R in the set A={1,2,3,4,5,6} as R={(x,y):y is divisible by x} (ii) Relation R in the set Z of all integers defined as R={(x,y):x−y is an integer}	04	Understand	CO2
	(c) Give : (i) A direct proof (ii) An indirect proof and (iii) Proof by contradiction, for the following statement: "If n is an odd integer, then (n + 9) is an even integer".	06	Apply	CO1

Course Outcomes meant to be assessed:

CO1: Apply the notion of mathematical logic & proofs in problem solving.
CO2: Solve problems which involve discrete data structures such as relations

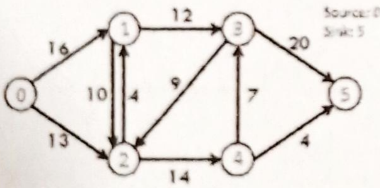
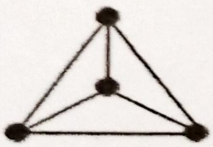
CIE: Internal Assessment Details

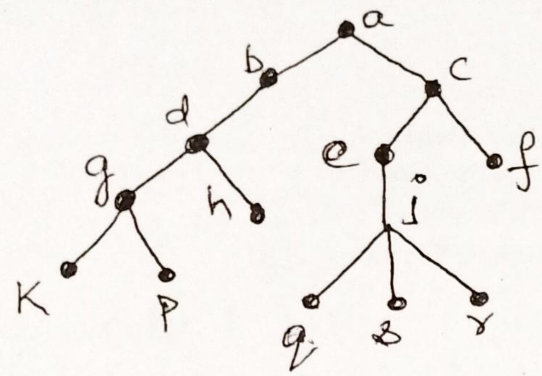
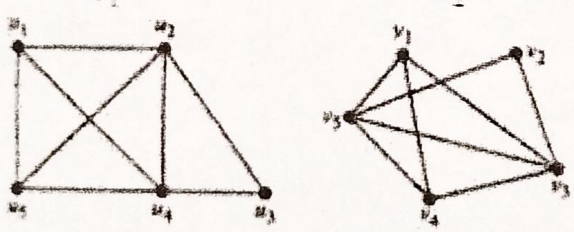
Internal Assessment Question Paper – II
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

Programme: B.E (CSE)	**Term:** Nov 2023 – Mar 2024	**Max Marks:** 30
CIE: Test 2	**Subject:** Discrete Mathematical Structures	**Sub Code:** CS35
Credits : 3:0:0	**Sem:** III	**Sec:** A, B, C
Date: 28/2/2024	**Time:** 10 - 11 AM	**Portions for Test:** (L11-L28)

Instructions to Candidate:

1. **Question 1 is Compulsory.** Answer any one question from 2 and 3.
2. Each Question carries 15M.
3. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl. No	Question	Marks	Bloom's Level	CO Mapping
1 a	Find the number of arrangements of the letters of the word INDEPENDENCE. In how many of these arrangements, (i) Do the words start with P? (ii) Do the vowels never occur together? (iii) Do the words begin with I and end in P?	04	Apply	CO3
b	Define Planar Graph .Verify whether Peterson Graph is planar or not.	06	Apply	CO4
c	Explain Ford-Fulkerson Algorithm for Maximum Flow Problem. Find the Maximum flow in the following graph.  	05 23	Apply	CO5
2 a	A certain question paper contains three parts A, B, C with four questions in part A, five questions in part B and six questions in part C. It is required to answer seven questions in selecting at least two questions from each part. In how many ways can a student select his seven questions for answering?	05	Apply	CO3
b	Define chromatic number of graph. Find the chromatic polynomial and if five colors are available how many possible ways are there to color the graph. 	05	Apply	CO4
c	(i) Define a tree. Prove that in every tree T = (V, E), |V| = |E| + 1 (ii) Answer the following questions for the tree shown below	05	Apply	CO5

	 i. Which vertices are the leaves? ii. Which vertex is the parent of g? iii. Which vertices are the descendants of c? iv. Which vertices are the siblings of s? v. What is the level of the vertex f? vi. Height of the tree			
3 a	Construct an optimal prefix code for the message ROAD IS GOOD (Considering space as a letter).	05	Apply	CO5
b	How many solutions are there to the equation x1 + x2 + x3 + x4 = 17, Where x1, x2, x3 and x4 are nonnegative integers?	05 1140	Apply	CO3
c	Define isomorphism of graphs. Examine whether the following graphs are isomorphic or not. G_1 G_2 	05 rat	Apply	CO4

Course Outcomes meant to be assessed:

CO3 : Apply basic counting techniques and combinatorics in the context of discrete probability.
CO4 : Demonstrate knowledge of fundamental concepts in graphs.
CO5 : Demonstrate knowledge of trees and its properties using various modelling techniques.

Internal Assessment Question Paper – 1

M.S. Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of CSE

Programme: B.E
Course: Digital Design and Computer Organization
Sem: III **CIE:** I
Term: Nov to Mar 2024
Course Code: CS32
Section: A, B and C

Max Marks: 30 **Time:** 1Hr **Portions for Test:** L1-L16.

Instructions to Candidates: Mobiles, smart watches or any electronic gadgets are strictly banned.
1st question is compulsory. Answer any **one** from Question 2 or Question 3.

Sl #		Question	Marks	Bloom's Level	CO Mapping
1	a)	Simplify the following function using K-Map, obtain the minimum SOP solution and realize the circuit using NAND gates only. Y=F(A,B,C,D) = $\sum m(0, 2, 3, 4, 5, 8, 10, 13, 14, 15)$	6	Apply	CO1
	b)	Derive the characteristic equation for SR-FF and T-FF.	4	Understand	CO2
	c)	Explain Binary to Gray code converter	5	Understand	CO1
2	a)	Determine the Essential Prime Implicant using Tabular Method for the given function. Y=F(A,B,C) = $\sum m(0,2,4,5,6)$	6	Apply	CO1
	b)	Draw the logic diagram of the JK flip-flop and explain the working with a suitable timing diagram.	5	Understand	CO2
	c)	Explain the working of 2:4 decoder.	4	Understand	CO1
3	a)	Implement the following function using a 4:1 Multiplexer using the appropriate method. Y=F(A,B,C) = $\sum m(1,2,6,7)$	4	Apply	CO1
	b)	Write the excitation table of D-FF, JK-FF and T-FF (or) Write the truth table of D-FF, JK-FF and T-FF	6	Understand	CO2
	c)	Explain the working of SR-Latch using NOR gates. Draw the timing diagram for it.	5	Understand	CO2

Course Outcomes meant to be assessed by the IA Test-I:

- CO1: Apply various techniques to minimize Boolean functions of various digital combinational circuits.
- CO2: Design asynchronous and synchronous sequential circuits using flip flops

Internal Assessment Question Paper – 2

Ramaiah Institute of Technology

(Autonomous Institute, Affiliated to VTU)

Department of Computer Science and Engineering

Programme : B.E	**Term:** Nov 2023-Feb 2024	**Max Marks:** 30	
CIE: Test 2	**Subject:** Digital design and Computer organization	**Subject Code:** CS32	**Credits:** 3:0:1
Sem:3	**Sec:**A,B,C	**Date:** 26/02/2024	**Time:**10am-11am

Portions for Test: (L17-L37)

Instructions to Candidates:

1. Question 1 is compulsory. Answer any one from question 2 or 3. Each Question carries 15M.
2. Mobiles, smart watches or any electronic gadgets are strictly banned.

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Design Mod 8 Synchronous up counter using T flip flops.	L3	2	5
b	Show the connection of the memory to the processor. Write brief note about the same.	L2	5	5
c	i. What is overflow in integer arithmetic? ii. Multiply 18 * 13 using Carry save addition and Show the steps.	L2	3,4	5
2.a	Draw the big endian and little endian assignments of byte addressability. Give examples.	L2	3	5
b	Multiply -4 (multiplicand) * -3(Multiplier) using booth algorithm and Show the steps. Mention the advantage of using booth algorithm.	L3	4	5
c	List the types of hazards in pipelining. Explain any one type of hazard in detail.	L2	4	5
3.a	How are the addressing modes used for the following methods of accessing operands? i. Indirection and pointers ii. Indexing and arrays	L3	3	5
b	What are the different stages in pipelining? Explain with neat diagram.	L2	3	5
c	Explain three bus organization of the datapath in basic processing unit with a diagram.	L2	4	5

* **L1** – Remember, **L2** – Understand, **L3**- Apply, **L4**- Analyze, **L5**-Evaluate, **L6**-Create

COs Mapped: 2. Examine the characteristics of flip flop sand design asynchronous and synchronous sequential circuits;3. Demonstrate basic functional units, instruction sequencing and various addressing modes; 4. Appraise different techniques used to perform multiplication and describe the organization of processing unit & pipelining; 5. Illustrate cache memory mapping techniques, data transfer between the devices using interrupts and direct memory access mechanism

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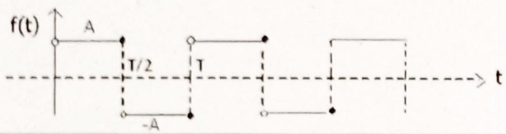
DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AI/AD/CI/CY31	**Sub:**	Laplace Transforms and Vector space	**Test:**	01
Semester:	III	**Term:**	20-11-2023 to 02-03-2024	**Marks:**	30
Date:	04.01.2024	**Time:**	03.30 – 04.30 PM		

Note: Answer any TWO full questions. Each main question carries 15 marks

Q.No.		Questions	Blooms Level	CO's	Marks
1.	(a)	Write the formula for $L\left(t\int_0^t f(t)dt\right)$	L1	CO1	2
	(b)	Given $L\{Sin\sqrt{t}\}=\frac{\sqrt{\pi}}{2s^{3/2}}e^{-1/4s}$, then find $L\left\{\frac{\cos\sqrt{t}}{\sqrt{t}}\right\}$	L2	CO1	3
	(c)	Evaluate i) $\int_0^\infty \frac{e^{-3t}\sin t}{t}dt$ (ii) $L\left\{t^9 e^{-\frac{3}{2}t}\right\}$	L3	CO1	5
	(d)	Obtain the Laplace transform of the $f(t)$ given by figure 	L5	CO1	5
2.	(a)	Define the Dirac-Delta function and sketch its graph.	L1	CO2	2
	(b)	Evaluate $L^{-1}\left(s\log\left(\frac{s+4}{s-4}\right)\right)$	L2	CO2	3
	(c)	Express the function $f(t)=\begin{cases} t^2, & 0<t<2, \\ 4t, & 2<t<4, \\ 8, & t>4. \end{cases}$ in terms of the Heaviside function and hence find its Laplace transform.	L3	CO2	5
	(d)	A voltage Ee^{-at} is applied at $t=0$ to a circuit of inductance L and resistance R. Show that the current at time $t=0$ is $\frac{E}{R-al}\left(e^{-at}-e^{-Rt/L}\right)$ using Laplace transforms.	L4	CO2	5
3.	(a)	Define a periodic function with an example.	L1	CO1	2
	(b)	Find the inverse Laplace transform of $\frac{1}{s(s^2+1)}$	L2	CO2	3
	(c)	If $L\{f(t)\}=F(s)$ then prove that $L\{t^n f(t)\}=(-1)^n\frac{d^n}{ds^n}\{F(s)\}$, where n is a positive integer.	L3	CO1	5
	(d)	Using Laplace transform, solve the simultaneous equation $\frac{dx}{dt}-2y=\cos 2t;\ \frac{dy}{dt}+2x=\sin 2t$, given $x(0)=1,\ y(0)=0.$	L4	CO2	5

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Mobiles are Banned

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DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AI/AD/CI/CY31	**Sub:**	Laplace Transform and Vector Space	**Section:**	CSE & Allied Branches
Semester:	III	**Term:**	20.11.2023 to 02.03.2024	**Marks:**	30
Date:	26-02-2024	**Time:**	3:00 PM-4:00 PM	**Test:**	II

Note: Answer any TWO full questions. Each main question carries 15 marks

Q. No.		Questions	Blooms Level	CO's	Marks
1.	(a)	Define basis and dimension of a vector space V.	L1	CO3	2
	(b)	Find the Eigen vector corresponding to the Eigen value $\lambda=3$ of the matrix $A=\begin{bmatrix}1&1&3\\1&5&1\\3&1&1\end{bmatrix}$.	L2 1 -1 1	CO5	3
	(c)	Determine the matrix that defines a reflection in the y-axis followed by rotation through an angle $\pi/3$ and then by the dilation of factor 7/2 . Find the image of the point $\begin{bmatrix}5\\2\end{bmatrix}$ under this transformation.	L3	CO3	5
	(d)	Construct orthonormal basis for the following using Gram-Schmidt process $x_1=\begin{bmatrix}1\\1\\1\end{bmatrix}, x_2=\begin{bmatrix}-1\\0\\1\end{bmatrix}, x_3=\begin{bmatrix}-1\\2\\3\end{bmatrix}, w=span\{x_1, x_2, x_2\}\in R^3$.	L4	CO4	5
2.	(a)	Write any two properties of Eigen values.	L1	CO5	2
	(b)	Prove that the transformation T: $R^2 \rightarrow R^2$ given by T(x, y)=(x−2y, 4x) is linear.	L2	CO3	3
	(c)	Find the least-squares solution of $Ax=b$, where $A=\begin{bmatrix}4&0\\0&2\\1&1\end{bmatrix}$ and $b=\begin{bmatrix}2\\0\\11\end{bmatrix}$.	L3 84 168	CO4	5
	(d)	Consider the bases B = {(1, 2), (3, -1)} and B' = {(3, 1), (5, 2)} of R^2. Find the transition matrix from B to B'. If u is a vector such that $u_B=\begin{bmatrix}3\\2\end{bmatrix}$, find $u_{B'}$.	L4 -2 3	CO3	5
3.	(a)	Define orthogonal basis with an example.	L1	CO4	2
	(b)	Find the null space of the matrix $A=\begin{bmatrix}1&2&3&5\\2&4&8&12\\3&6&7&13\end{bmatrix}$.	L2	CO4	3
	(c)	Determine the kernel and range of each of the transformation T(x, y) = (5x, x+y, 2y) of $R^2 \rightarrow R^3$ and hence verify Rank nullity theorem.	L3	CO3	5
	(d)	Diagonalize the matrix $A=\begin{bmatrix}3&1\\1&3\end{bmatrix}$.	L5	CO5	5

Quiz Test

Sub: **Laplace Transforms and Vector Space**

Sub. Code: **CS/IS/AI/AD/CI/CY31**

Date: **21/02/2024**

Time: **4.45 – 5.00 pm**

1.	Evaluate $L\left\{t\int_0^t \frac{\sin t}{t}\,dt\right\}$.
2.	Find $L\{(t^2-6t+9)\,e^{-t}\,U(t-3)\}$.
3.	Determine the matrix that defines a reflection in the x-axis followed by rotation through an angle $\pi/3$.
4.	Construct the orthonormal basis of the subspace spanned by $\left\{\begin{bmatrix}1\\2\end{bmatrix},\begin{bmatrix}2\\-3\end{bmatrix}\right\}$ of R^2.
5.	Write any two properties of Eigen values and Eigen vectors.

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Internal Assessment Question Paper Programme: B.E		
Semester: III	Course Name: Unix Shell Programming	Course Code: CSAEC310
CIE: Test 1	Term: 2023-24	Section: A, B & C
Max. Marks: 30	Duration of Test: 1 Hour	Time: 3.30 to 4.30
Credits: 1:0:0	Test Portions: L1-L6	Date:03.01.2024

Note: Mobiles and Programmable Calculators are strictly prohibited

Instructions To Candidates:

- Section I is compulsory
- Answer Two Full Questions from Section II

Section-I
Objective type One-Mark Questions

Answer the following Questions: **6 X 1 = 6Marks**

SL. No.	Questions	CO
1	Which command is used for extracting the details of the operating system? a) cd b) echo c) uname d) wc	1
2	What is the output of who command? a) display information about users who are currently logged in. b) display file hierarchy c) display administrator information d) display processes	1
3	A file named abd.txt has the following set of permissions -rwxrwxrwx All the three operations i.e read, write and execute can be performed on the file by file owner, group owner and others. a) True b) False	2
4	Which filter is used in UNIX to translate characters from standard input? a)tr b) tee c) cut d) head	2
5	_____ command will bring the background jobs to the foreground. a) bg b) fg c) ctrl-Z d) kill	2
6	What is the output of $ wc -c *.c command? (a)size of each .c file b) total size of .c files c) erroneous d) undefined	2

Section-II

Answer any TWO full questions **12 X 2 = 24 Marks**

SL. No.		Questions	Marks	BL	CO
1	(a)	Explain the architecture of UNIX in detail.	3	L2	1
	(b)	Briefly explain the following commands: (i) wc (ii) man (iii)tar	3	L2	2
	(c)	Explain 'disk utilities' command with different options.	6	L2	2
2	(a)	Define **Vi** Editor and Explain its modes.	3	L2	1
	(b)	Write a short note on file permissions	3	L2	2
	(c)	List and explain networking commands.	6	L2	2
3	(a)	Explain **'more'** command with different options	3	L2	1
	(b)	Briefly explain the following file changing permission modes: Symbolic Mode &Absolute Mode	3	L2	2
	(c)	List and explain UNIX commands used for text processing.	6	L2	2

CO1: Elaborate the structure of UNIX environment and basic commands.
CO2. Discuss the interpretive nature of the shell, Unix Process, File structure, Directories and their associated system calls.

Internal Assessment Question Paper Programme: B.E		
Semester: III	Course Name: Unix Shell Programming	Course Code: CSAEC310
CIE: Test 2	Term: 20.11.2023-2.03.2024	Section: All
Max. Marks: 30	Duration of Test: 1 Hour	Time: 3.00 – 4.10
Credits: 1:0:0	Test Portions: L07 to L14	Date:28.02.2023

Note: Mobiles and Programmable Calculators are strictly prohibited

Instructions To Candidates:

- Section I is compulsory
- Answer Two Full Questions From Section II

Section-I
Objective type One-Mark Questions

Answer the following Questions: **6 X 1 = 6Marks**

S. No.	Questions	CO
1	What does the "2>&1" mean in a command? **a)** Redirect stdout to a file **b)** Redirect stderr to stdout **c)** Redirect stdin to stdout **d)** Redirect stdout and stderr to a file	3
2	________________ provides a means of creating customized commands by assigning name to a command. a) redirection b) alias c) flag d) job	3
3	Which of the following is not filter in unix? a) cat b) head c) cd d) tail	4
4	What command is used to extract specific columns from a text file? a)cut b) paste c) extract d) split	4
5	Which of the following does not belong to family of grep? a) egrep b) fgrep c) mgrep d) grep	5
6	Find any error in each of the following sed instructions? a) 25d b) 30,50,70d c) 20,s/A/B/g d) 25,/^A. *A$/s/A//g	5

Section-II

Answer any TWO full questions **12 X 2 = 24 Marks**

SL. No.		Questions	Marks	BL	CO
1	(a)	Define redirection. With a relevant example, explain input and output redirection.	6	L2	3
	(b)	Name the command that UNIX provides to concatenate files with example	2	L2	4
	(c)	Define regular expressions. Illustrate atoms and operators with the help of suitable examples.	4	L2	5
2	(a)	With a neat diagram, explain the working of pipe.	5	L2	3
	(b)	Briefly describe the head and tail commands and list the available options.	4	L3	4
	(c)	What does grep stand for? Draw a grep operation flow chart.	3	L2	5
3	(a)	Explain in detail three standard streams of UNIX.	5	L2	3
	(b)	Compare the contents of two files by examining how the cmp, diff, and comm commands operate.	3	L2	4
	(c)	Define sed command and draw the diagram of sed command.	4	L1	5

CO3: Develop Shell programs with customization of environmental variables.
CO4: Design a shell script for specific tasks using filters and pipes.
CO5: Discuss the use of regular expression in various types of filtering

24/2/24
Co-ordinator

24/2/2024
HOD, CSE

RAMAIAH
Institute of Technology

Continues Internal EXAMINATIONS – Jan 2024

Program	:	B.E. : All the Branch	Semester	:	III
Course Name	:	Universal Human Values	Max. Marks	:	50
Course Code	:	UHV38	Duration	:	1 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Write figures wherever necessary.

TERM :20th NOV to 3rd MAR 2024 **PORTIONS : L1-L14**

DATE : 05-01-2024 **TIME: 11.00AM-12.00**

Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.** Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Determine the extent to which value education is essential to society's professional development.	L2	CO1	7
b	What are the prevailing notions of Happiness and Prosperity? Explain How the false notions of happiness and Prosperity have affected human living at all four levels.	L1	CO1	8
2.a	Illustrate the three levels of understanding the Activities in the Self and the Activities in the Body with examples.	L2	CO2	7
b	Explain the Self "I" and body in the form of Sanyama and Svasthya and illustrate the programs to ensure them.	L2	CO2	8
3.a	State the terms svatva, swatantrata and swarajya.	L1	CO1	3
b	State the requirements needed to fulfill the human aspirations.	L2	CO1	4
c	Discuss how realization and understanding are essential for happiness and harmony.	L2	CO2	8

*L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create

Ramaiah Institute of Technology, Bengaluru – 560 054
(Autonomous Institute, Affiliated to VTU)
Programme: B. E. (Common to all branches)

Term:	20.11.2023 – 02.03.2024	Course Code:	UHV38
Course:	Universal Human Values	Semester:	III
CIE:	Test – II	Date:	27.02.2024
Max. Marks	30	Time:	10:00-11:00AM

Note: Mobiles and programmable calculators are banned strictly

Answer any two questions

1	a	Do human values affect the life of professionals? Justify your answer with two examples.	7	CO5
	b	Explain the characteristics of four orders of nature.	8	CO4
2	a	What is the meaning of respect? How do we disrespect others due to lack of right understanding of this feeling?	7	CO3
	b	What do you mean by sustainable development? Discuss.	8	CO5
3	a	Explain how production activities can be enriching to all the orders of nature. Give any two examples.	7	CO3
	b	What do you mean by 'conformance'? Explain the conformance in the four orders.	8	CO4

CIE: Internal Assessment Details

Internal Assessment Question Paper – 1
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

Programme : B.E **Term: Nov 2023– Mar 2024** **Max Marks:** 30
CIE: Retest **Subject:**Data Structures **Subject Code: CS33** **Credits: 3:0:0**
Sem: 3 **Sec: A,B,C** **Date:** **27/01/2024** **Time:9.30AM-10.30AM**
Portions for Test: (L1-L20)

Instructions to Candidates:

1. **Question 1 is compulsory**. Answer any one from question 2 or 3. Each Question carries 15M.
2. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl#		Question	Marks	Bloom's Level	CO Mapping
1	a.	Justify the statement "Malloc may be invoked from several places in your program, it is often convenient to define the macro that invokes malloc to create a memory and exits when malloc fails" with a code snippet.	5	L3	1
	b.	Write a C function for fast transpose of the sparse matrix.	5	L2	2
	c.	Write a C function to insert a node at the rear end of the list.	5	L2	3
2	a.	What is dangling reference? Explain with a code snippet.	4	L2	1
	b.	Write a recursive C function to search for a given key element using binary search algorithm.	5	L2	1
	c.	Write C function to convert from infix to postfix expression.	6	L2	2
		OR			
3	a.	Apply infix to postfix conversion algorithm to convert the given infix expression to postfix expression (Represent the operations step-by-step by showing in a tabular form, how stacking and unstacking is done). $(a + b) * d + e / (f + a * d) + c$	5	L3	2
	b.	Write C function for the evaluation of postfix expression.	6	L2	2
	c.	Justify the statement "pointers can be dangerous" with a code snippet.	4	L3	1

Course Outcomes meant to be assessed by the IA Retest:

1. Solve real time problems using concepts of dynamic memory allocation, structures, and strings. (PO-1,2,3, PSO-2)
2. Implement storage and retrieval of ordered data using stacks and queues as well as select appropriate data structures as applied to specified problem definition (PO-1,2,3, PSO-2)
3. Implement dynamic storage, retrieval and search operations of unordered data using linked list and its variants. (PO-1,2,3, PSO-2)

CIE: Internal Assessment Details

Internal Assessment Question Paper – 2
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

Programme : B.E **Term: Nov 2023– Mar 2024** **Max Marks:** 30
CIE: Test 2 **Subject:**Data Structures **Subject Code: CS33** **Credits: 3:0:0**
Sem:3 **Sec:A,B,C** **Date: 29/2/2024** **Time:10-11am**
Portions for Test: (L21-L42)
Instructions to Candidates:

1. **Question 1 is compulsory**. Answer any one from question 2 or 3. Each Question carries 15M.
2. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl#		Question	Marks	Bloom's Level	CO Mapping
1	a.	Explain linked list representation of sparse matrix and write C code to represent its node structure.	5	L3	3
	b.	Construct a binary tree from its given Inorder and Preorder sequences Preorder sequence: ABCDEFGHI Inorder sequence: BCAEDGHFI	5	L3	4
	c.	With a C function, explain Depth First Search.	5	L3	5
2	a.	Illustrate with an example the polynomial representation using list.	5	L2	3
	b.	Differentiate between max and min heap. Write a C function to insert an element into a max heap.	5	L4	4
	c.	How do you calculate the balance factor of a node in an AVL tree? Construct AVL Tree for the nodes: 16, 27, 9, 11, 36, 54.	5	L3	5
		OR			
3	a.	What is a circular linked list? Write a C function to perform insertion of a node at the front of the list.	5	L3	3
	b.	What is Binary Search Tree? With the C functions explain Inorder, Preorder and Postorder traversal of a tree.	5	L2	4
	c.	Name the different types of leftist trees. Discuss each with an example.	5	L3	5

Course Outcomes meant to be assessed by the IA Test2:

3) Implement dynamic storage, retrieval and search operations of unordered data using linked list and its variants. (PO-1,2,3, PSO-2)
4) Implement hierarchical based solutions using different tree traversal techniques. (PO-1,2,3, PSO-2)
5) Develop solutions for problems based on graphs. (PO-1,2,3, PSO-2)